NIT6160: Data Warehousing and Mining

Final Project

Md Anwar Hossain

Student ID: 8233581

June 14, 2026

Abstract

Cardiovascular disease remains a major public health concern, and early identification of individuals at risk is important for prevention and clinical decision-making. As part of the class assignment, this study analysed a benchmark cardiovascular disease dataset obtained from Kaggle, consisting of demographic, clinical, anthropometric and lifestyle-related variables. The original dataset contained 70,000 patient records, and after preprocessing and removal of implausible observations, 66,364 records were retained for analysis. Age was converted from days to years, body mass index (BMI) was derived from height and weight, and selected predictors were standardised for models sensitive to feature scale. Four machine learning models were developed and compared: Logistic Regression, Decision Tree, Random Forest and Support Vector Machine. Model performance was evaluated using accuracy, precision, recall, F1-score and ROC-AUC. Random Forest achieved the strongest overall performance, with an accuracy of 0.734 and ROC-AUC of 0.799, although Logistic Regression performed almost equally well, with an accuracy of 0.732 and ROC-AUC of 0.797. Feature importance analysis showed that systolic blood pressure, diastolic blood pressure, age, cholesterol and BMI were the most influential predictors of cardiovascular disease. Lifestyle variables contributed less strongly, possibly due to their simplified binary representation in the dataset. Overall, the findings suggest that routine clinical and demographic variables can support useful cardiovascular disease prediction, and that interpretable models such as Logistic Regression may remain valuable when their performance is close to more complex models.

Table of Contents

List of Tables

List of Figures

1. Introduction

Cardiovascular disease (CVD) remains the leading cause of death worldwide and represents a major public health challenge. According to the World Health Organization, approximately 17.9 million people die from CVD each year, accounting for nearly one-third of all global deaths. Early identification of individuals at elevated risk is therefore essential for implementing preventive interventions and reducing disease burden. Traditional risk assessment approaches rely on clinical judgement and established risk factors such as age, blood pressure, cholesterol, obesity, smoking and diabetes. However, the growing availability of large health datasets has created opportunities to apply machine learning techniques that can identify complex relationships among risk factors and improve disease prediction.

The cardiovascular disease dataset used in this study contains 70,000 patient records and includes demographic variables (age and gender), clinical measurements (blood pressure, cholesterol and glucose levels), anthropometric characteristics (height, weight and body mass index), and lifestyle factors (smoking, alcohol consumption and physical activity). Previous studies have consistently identified hypertension, dyslipidaemia, obesity, ageing and hyperglycaemia as major contributors to cardiovascular risk [1, 2]. More recently, machine learning methods have been applied to cardiovascular datasets to improve risk prediction and identify influential predictors [3, 4]. Nevertheless, uncertainty remains regarding the extent to which relationships between cardiovascular risk factors and disease outcomes are linear or nonlinear, and whether more complex models provide meaningful improvements over simpler and more interpretable approaches. To address this issue, four classification methods representing different modelling philosophies were selected: Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), and Support Vector Machine (SVM).

Logistic Regression is a widely used statistical model for binary classification and estimates disease probability. The principal advantage of Logistic Regression is interpretability because model coefficients directly quantify the direction and magnitude of predictor effects. However, the model assumes a linear relationship between predictors and the log-odds of the outcome and may perform poorly when complex nonlinear relationships exist. *Decision Trees* provide a rule-based alternative by recursively partitioning the feature space into increasingly homogeneous groups. Decision Trees can automatically capture nonlinear effects and interactions among predictors and are highly interpretable. However, they are sensitive to small changes in the training data and are prone to overfitting when trees become overly complex. *Random Forest* extends the Decision Tree framework by constructing a large ensemble of trees from bootstrap samples and aggregating their predictions. By averaging predictions across multiple trees, Random Forest reduces variance and improves generalisation performance. It can effectively model complex interactions and nonlinear relationships but sacrifices interpretability because individual decision pathways are no longer easily traceable. *Support Vector Machine* (SVM) seeks an optimal separating hyperplane that maximises the margin between classes. Although SVM often achieves strong predictive performance, it requires feature scaling and parameter tuning, and its predictions are generally more difficult to interpret clinically.

Recent evidence suggests that ensemble and kernel-based methods frequently outperform traditional statistical models in cardiovascular risk prediction, although gains in predictive accuracy are often accompanied by reduced interpretability [4, 5]. In clinical applications, model transparency remains an important consideration because healthcare practitioners must understand the factors driving predictions before incorporating them into decision-making processes.

Based on the literature and characteristics of the selected dataset, this study addresses the following research questions: (i) Which demographic, clinical and lifestyle variables contribute most strongly to cardiovascular disease risk? (ii) Are the relationships between cardiovascular disease and predictor variables primarily linear, or do important nonlinear effects and interactions exist? (iii) How accurately can Logistic Regression, Decision Tree, Random Forest and Support Vector Machine models predict cardiovascular disease? (iv) Which machine learning model provides the best balance between predictive performance and interpretability? (v) What insights do the most influential predictors provide regarding cardiovascular disease prevention and risk management?

By combining exploratory data analysis, feature engineering and comparative machine learning modelling, this study aims to identify key cardiovascular risk factors and evaluate the suitability of different predictive approaches for healthcare decision support.

2. Literature Review

Cardiovascular disease (CVD) prediction has increasingly relied on machine learning because clinical, demographic and lifestyle variables often interact in complex ways. Bays et al. [1] emphasised that CVD risk is shaped by multiple modifiable and non-modifiable factors, including age, sex, blood pressure, dyslipidaemia, hyperglycaemia, obesity, smoking, diet and physical inactivity. Similarly, Powell-Wiley et al. [2] highlighted obesity as an important contributor to CVD through its association with hypertension, dyslipidaemia and diabetes, suggesting that body-size indicators such as BMI may provide useful predictive information.

Recent machine learning studies have shown that CVD can be predicted with reasonable accuracy using routine patient-level variables. Bhatt et al. [3] analysed a real-world dataset supplied by Svetlana Ulianova in Kaggle based on 70,000 observations for cardiovascular disease related to lifestyle and health related attributes using various computational models to infer comparative predictive performance for decision tree, random forest, XGBoost and multilayer perceptron models. Han [6] also used the same Kaggle dataset and found that systolic blood pressure, diastolic blood pressure, age and cholesterol were more influential predictors than smoking, height and gender. These findings support the use of feature-importance analysis in addition to model accuracy metrics.

Broader reviews also indicate that machine learning models can outperform conventional statistical approaches in CVD risk prediction, although issues such as interpretability, overfitting and external validation remain important. Liu et al. [4] and Al Jowf and Kolhar [5] further showed that logistic regression, random forest, gradient boosting and multilayer perceptron can each contribute differently to CVD prediction depending on preprocessing, feature engineering and evaluation metrics. However, support vector machines have been less consistently applied in studies using this specific dataset. Therefore, this study compares Logistic Regression, Decision Tree, Random Forest and Support Vector Machine models to evaluate predictive performance and identify the strongest contributors to cardiovascular disease risk.

3. Methods

3.1 Dataset Description

This study utilised the Cardiovascular Disease Dataset developed by Ulianova and publicly available through the Kaggle repository. The dataset contains information from 70,000 individuals and was originally collected to support research on cardiovascular disease risk prediction. Each observation represents a single patient and includes demographic characteristics, clinical measurements, and lifestyle-related variables commonly associated with cardiovascular health.

The dataset consists of 11 predictor variables and one binary target variable indicating the presence (1) or absence (0) of cardiovascular disease (Table 1). Predictor variables include age, gender, height, weight, systolic blood pressure (ap_hi), diastolic blood pressure (ap_lo), cholesterol level, glucose level, smoking status, alcohol consumption, and physical activity. Age was originally recorded in days, while cholesterol and glucose were represented using ordinal categories indicating normal, above-normal, and well-above-normal levels.

The target variable was approximately balanced, with nearly equal numbers of individuals diagnosed with cardiovascular disease and individuals without disease. Such class balance reduces the risk of biased model training and allows conventional classification performance metrics to provide reliable estimates of model performance.

Table 1. Description of variables included in the cardiovascular disease dataset.

Variable	Description	Type	Values/Unit
age_years	Age of participant (converted from days to years)	Continuous	Years
gender	Sex of participant	Categorical	1 = Female, 2 = Male
height	Height of participant	Continuous	cm
weight	Body weight of participant	Continuous	kg
BMI	Body Mass Index calculated as weight (kg) divided by height squared (m^2)	Continuous	kg/m^2
Ap_hi	Systolic blood pressure	Continuous	mmHg
Ap_lo	Diastolic blood pressure	Continuous	mmHg
cholesterol	Cholesterol level	Ordinal	1 = normal, 2 = Above normal, 3 = Well above normal
gluc	Blood glucose level	Ordinal	1 = normal, 2 = Above normal, 3 = Well above normal
smoke	Smoking status	Binary	0 = No, 1 = Yes
alco	Alcohol consumption	Binary	0 = No, 1 = Yes
active	Physical activity status	Binary	0 = inactive, 1 = active
cardio	Cardiovascular disease status (target variable)	Binary	0 = no disease, 1 = disease

The original dataset contained 70,000 observations and 12 variables, including 11 predictor variables and one target variable (cardio). During preprocessing, age was converted from days to years and Body Mass Index (BMI) was derived from height and weight to better represent obesity-related cardiovascular risk. Following outlier removal and data cleaning procedures, the final analytical dataset consisted of 66,364 observations.

3.2 Data Preprocessing and Feature Engineering

Data preprocessing was undertaken to improve data quality and ensure that predictor variables were suitable for statistical analysis and machine learning modelling. The age variable was converted from days to years to improve interpretability. In addition, Body Mass Index (BMI) was calculated using the standard equation: $BMI =$

$\frac{weight\ (kg)}{Height\ (m)^2}$. BMI was included because obesity is a well-established cardiovascular risk factor and is often more informative than body weight alone.

A series of data quality checks identified implausible observations and potential measurement errors. For age, height, and weight, outliers were detected using the interquartile range (IQR) method, where observations outside 1.5 times the interquartile range were removed. Clinical variables were further screened using medically realistic thresholds. Systolic blood pressure values were restricted to 80–250 mmHg, diastolic blood pressure values to 40–150 mmHg, and BMI values to 15–60 kg/m². These thresholds were selected to exclude physiologically unrealistic records while retaining genuine variation within the population.

Following data cleaning and outlier removal, the dataset size was reduced from 70,000 to 66,364 observations. The resulting dataset retained sufficient statistical power while providing improved data reliability for model development.

3.3 Assessment of Multicollinearity

Prior to model construction, relationships among predictor variables were examined using Pearson correlation coefficients. The correlation matrix revealed a strong positive correlation between BMI and body weight (r = 0.85). Because BMI is derived directly from height and weight and is more clinically meaningful for assessing obesity-related health risks, body weight was excluded from subsequent modelling to minimise redundancy.

Moderate correlation was observed between systolic and diastolic blood pressure (r = 0.72). However, both variables were retained because they represent different physiological dimensions of cardiovascular function and are routinely considered independent risk indicators in clinical practice.

3.4 Model Development and Training Strategy

The cleaned dataset was divided into training (80%) and testing (20%) subsets using stratified random sampling to preserve the proportion of cardiovascular disease cases in each subset. Model development was conducted exclusively on the training dataset, while the testing dataset was reserved for final model evaluation.

Because Logistic Regression and Support Vector Machines are sensitive to variable scale, continuous predictors were standardised using z-score normalisation:

$$Z = \frac{X - \mu}{\sigma}$$

where X is the original value, μ is the feature mean, and σ is the standard deviation. The standardization was done using the following code implementation from sklearn library:

```python
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Standardisation ensures that predictors contribute equally during optimisation and prevents variables with larger numerical ranges from dominating model estimation. Scaling was not required for Decision Tree and Random Forest models because tree-based algorithms are invariant to monotonic transformations of predictor variables.

3.5 Logistic Regression

Logistic Regression was selected as a baseline model because of its simplicity and interpretability. The probability of cardiovascular disease was estimated using the logistic function:

$$P(Y = 1|X) = \frac{1}{1+e^{[-(\beta_0+\sum_{i=1}^{n}\beta_i X_i)]}}$$

where $P(Y = 1|X)$ = probability that an individual has cardiovascular disease, β_0 = intercept, β_i = regression coefficient for predictor X_i, and X_i = predictor variables (e.g., age, BMI, blood pressure, cholesterol).

To allow the model to capture potential non-linear effects and interactions among predictors, second-order polynomial features were generated. These included quadratic terms (e.g., Age^2 and BMI^2) and interaction terms (e.g., Age × Blood Pressure). Regularised Logistic Regression was then applied to reduce overfitting associated with the expanded feature space.

3.6 Decision Tree

Decision Trees were employed to model non-linear relationships and hierarchical interactions among cardiovascular risk factors. The algorithm recursively partitions the predictor space into increasingly homogeneous subgroups using a sequence of decision rules that maximise class separation. The modelling process can be represented as:

```
Root Node
  |
  ├── Rule 1 (e.g., BP > 140)
  |       ├── Yes
  |       └── No
  |
  └── Further Splits
          ↓
     Disease / No Disease
```

3.7 Random Forest

Random Forest was implemented as an ensemble extension of the Decision Tree algorithm. The method constructs multiple trees using bootstrap samples of the training data and combines their predictions through majority voting. The final prediction can be expressed as:

$$\hat{Y} = \frac{1}{T} \sum_{t=1}^{T} h_t(X)$$

where, $\hat{Y}$ = final Random Forest prediction, T = total number of trees in the forest, $h_t(X)$ = prediction from the t-th decision tree for observation X.

3.8 Support Vector Machine

Support Vector Machine (SVM) classification was used to identify an optimal separating hyperplane between cardiovascular disease and non-disease cases. For non-linearly separable data, the radial basis function (RBF) kernel was employed to project observations into a higher-dimensional feature space.

The classification function can be expressed as:

$$f(x) = sign(w^T x + b)$$

where, w= weight vector (normal to the hyperplane), x= feature vector, $w^T x$= dot product between the weights and features, b= bias (intercept) term, and the $sign$ function determines the predicted class (e.g., cardiovascular disease or no-cardiovascular disease) based on which side of the separating hyperplane the observation falls.

For datasets exhibiting nonlinear patterns, kernel functions can project observations into higher-dimensional feature spaces, allowing more flexible decision boundaries. In this study, a radial basis function (RBF) kernel was used because of its ability to capture complex relationships among cardiovascular risk factors.

3.9 Hyperparameter Tuning Strategy

Hyperparameter tuning was performed to improve model generalisation and reduce overfitting. For Logistic Regression, Decision Tree and Random Forest, Grid Search Cross-Validation was applied using 10-fold cross-validation. Model selection was based on the mean cross-validated ROC-AUC score, as this metric evaluates the ability of each model to distinguish between cardiovascular disease and non-disease cases across different classification thresholds. For Support Vector Machine, Randomized Search Cross-Validation was used instead of exhaustive grid search because SVM training is computationally more expensive on a large dataset.

For Logistic Regression, the regularisation penalty type and regularisation strength parameter (C) were tuned. The optimal model used L2 regularisation with (C = 0.01).

For the Decision Tree model, maximum tree depth, minimum samples required for node splitting, and minimum samples per leaf node were tuned. The final model selected a maximum depth of 5 and a minimum leaf size of 10 observations.

For Random Forest, the number of trees, maximum tree depth, minimum samples required for splitting, minimum samples per leaf node, and number of features considered at each split were tuned. The optimal model used 200 trees, a maximum depth of 10, a minimum leaf size of 5, and square-root feature selection.

For Support Vector Machine, the regularisation parameter (C), kernel type, and kernel coefficient (γ) were tuned using Randomized Search Cross-Validation. The final selected model used an RBF kernel, (C = 1), and (γ= 0.01).

3.10 Model Evaluation

Model performance was evaluated using the independent testing dataset. Multiple evaluation metrics were calculated to provide a comprehensive assessment of predictive performance, including accuracy, precision, recall, F1-score, and the area under the receiver operating characteristic curve (ROC-AUC).

Accuracy measures the overall proportion of correctly classified observations, whereas precision quantifies the proportion of predicted positive cases that are truly positive. Recall evaluates the ability of a model to correctly identify individuals with cardiovascular disease, while the F1-score provides a balanced measure of precision and recall. ROC-AUC was included because it evaluates model discrimination across all classification thresholds and is widely regarded as one of the most informative measures for binary classification problems.

In addition to these metrics, confusion matrices were used to examine classification errors, and ROC curves were generated to visualise the trade-off between sensitivity and specificity for each modelling approach. The final model comparison considered both predictive performance and interpretability, recognising the importance of transparent decision-making in healthcare applications.

4. Results

4.1 Exploratory Data Analysis

After preprocessing and outlier removal, the final analytical dataset contained 66,364 observations. The original dataset contained no missing values, which meant that imputation was not required. The main preprocessing changes were the conversion of age from days to years, the calculation of body mass index (BMI), and the removal of physiologically implausible values in height, weight, blood pressure and BMI.

The distribution plots showed that most continuous variables followed plausible population-level patterns after cleaning. Age was concentrated mainly between approximately 40 and 65 years, while height and weight were centred around typical adult values. BMI (representing height and weight) showed a right-skewed distribution, indicating that a smaller proportion of individuals had substantially higher body mass. Systolic (ap_high) and diastolic (ap_low) blood pressure values were also right-skewed, with most observations falling within normal to moderately elevated ranges and fewer individuals showing very high blood pressure values (Fig. 1).

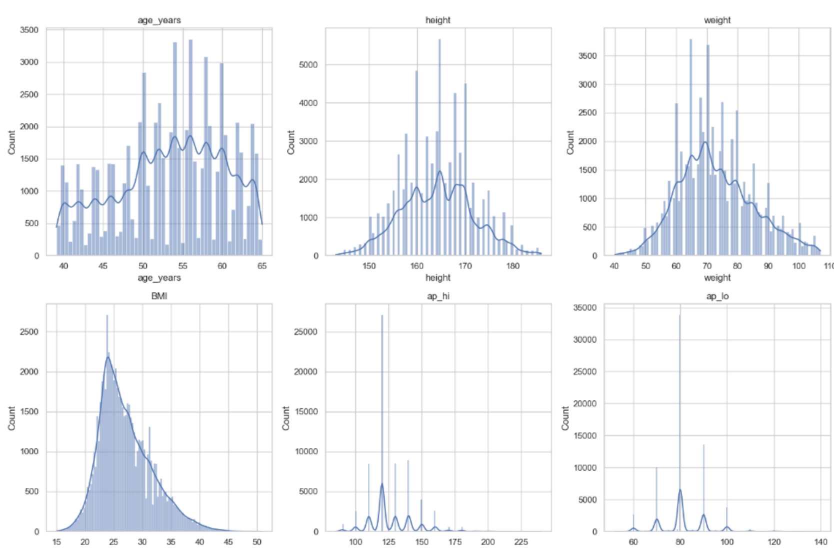

Figure 1. Distribution of continuous variables after preprocessing.

The target variable was approximately balanced, with similar numbers of individuals classified as having cardiovascular disease and not having cardiovascular disease (Fig. 2). This balance was useful for model development because it reduced the likelihood that models would favour one class simply because it was more common.

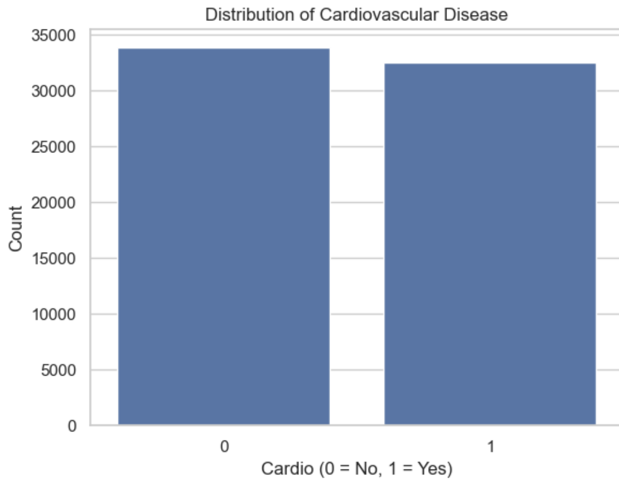

Figure 2. Distribution of cardiovascular disease target variable.

The boxplots comparing continuous variables by disease status showed clear differences between patients with and without cardiovascular disease. Individuals with cardiovascular disease generally had higher age, BMI, systolic blood pressure and diastolic blood pressure values. The separation was particularly visible for systolic blood pressure, where the disease group showed a higher median and wider upper range. This pattern suggested that blood pressure was likely to be an important predictor in the classification models.

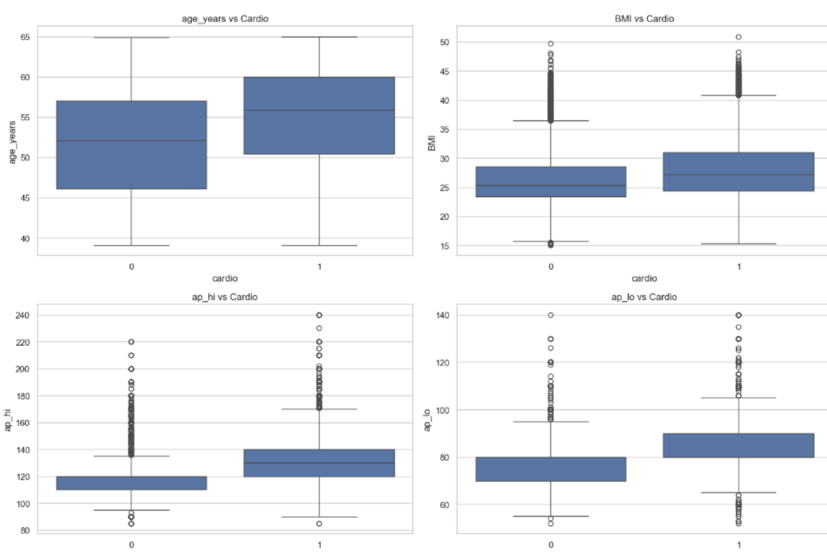

Figure 3. Boxplots of age, BMI, systolic and diastolic blood pressure by disease status.

The categorical plots also showed meaningful variation across disease status. Individuals with above-normal or well-above-normal cholesterol levels appeared more frequently in the cardiovascular disease group than in the non-disease group. Glucose showed a similar, although weaker, pattern. In contrast, smoking, alcohol consumption and physical activity showed relatively small differences between disease and non-disease groups in the visual summaries (Fig. 4).

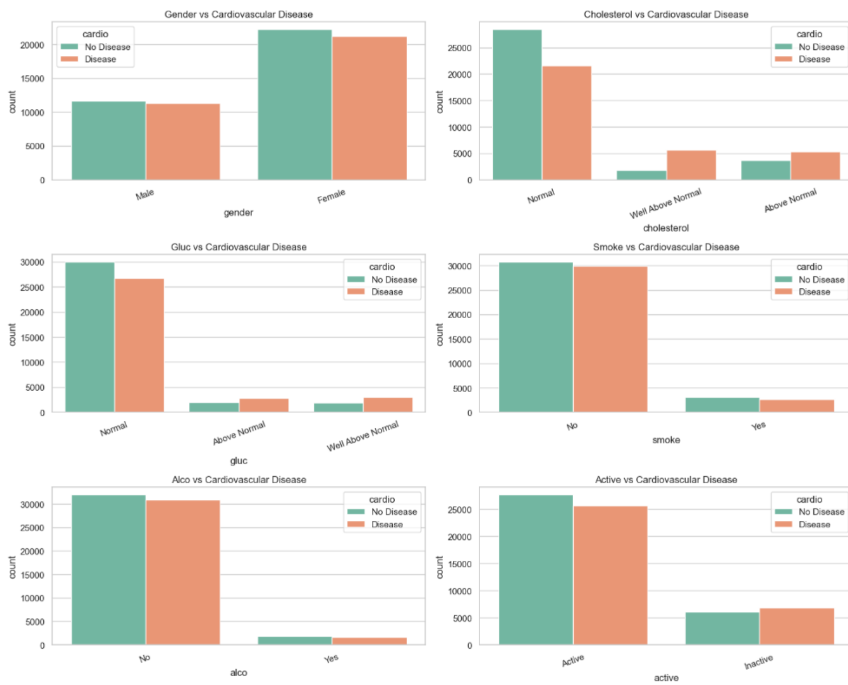

Figure 4. Categorical variables by cardiovascular disease status.

The correlation matrix supported these exploratory findings (Fig. 5). Systolic blood pressure had the strongest correlation with cardiovascular disease, followed by diastolic blood pressure, age, cholesterol and BMI. These variables were therefore expected to contribute strongly to model prediction. The correlation matrix also showed high correlation between BMI and weight, supporting the decision to remove weight from the modelling stage while retaining BMI as the more clinically interpretable obesity-related variable.

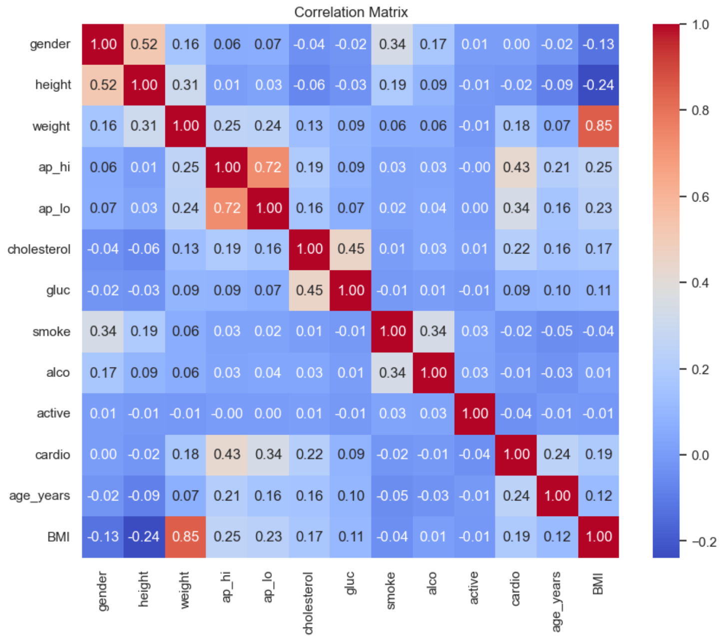

Figure 5. Correlation matrix of predictor variables.

4.2 Model Performance Comparison

Four classification models were evaluated on the hold-out test dataset: Logistic Regression, Decision Tree, Random Forest and Support Vector Machine. Model performance was assessed using accuracy, precision, recall, F1-score and AUC-ROC.

Table 2. Comparative performance of the four machine learning models. The best performing models were highlighted.

	Model	Accuracy	Precision	Recall	F1	ROC-AUC
0	Logistic Regression	0.732088	0.738793	0.701552	0.719691	0.796866
1	Decision Tree	0.729451	0.751640	0.669279	0.708073	0.791171
2	Random Forest	0.733896	0.756953	0.673429	0.712752	0.799346
3	SVM	0.730280	0.756260	0.663747	0.706990	0.791403

Overall, all four models achieved similar predictive performance, with accuracy values ranging from approximately 0.729 to 0.734. Random Forest produced the highest accuracy, precision and AUC-ROC, although the improvement over Logistic Regression was small. Logistic Regression achieved the highest recall and F1-score, indicating that it was slightly better at identifying actual cardiovascular disease cases than the other models. Decision Tree and SVM performed competitively but had lower recall, meaning they missed a larger number of disease cases.

4.3 Logistic Regression

The Logistic Regression model achieved an accuracy of 0.732, precision of 0.739, recall of 0.702, F1-score of 0.720 and AUC-ROC of 0.797 (Table 2). The confusion matrix showed 5,152 true negatives and 4,565 true positives (Fig. 6). However, the model also produced 1,614 false positives and 1,942 false negatives. This indicates that while the model performed reasonably well overall, some patients with cardiovascular disease were still incorrectly classified as non-disease cases.

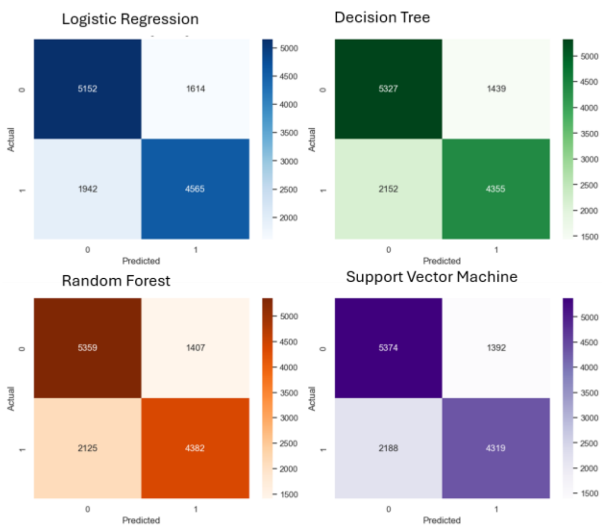

Figure 6. Confusion matrices for the selected models.

The ROC curve for Logistic Regression remained well above the diagonal reference line, showing good discrimination between disease and non-disease cases (Fig. 7). At lower false-positive rates, the model still identified a substantial proportion of true disease cases. For example, at approximately 20% false-positive rate, the curve indicated that the model detected about 65% of cardiovascular disease cases. This supports the AUC-ROC value of 0.797 and suggests that Logistic Regression provided a useful baseline classifier.

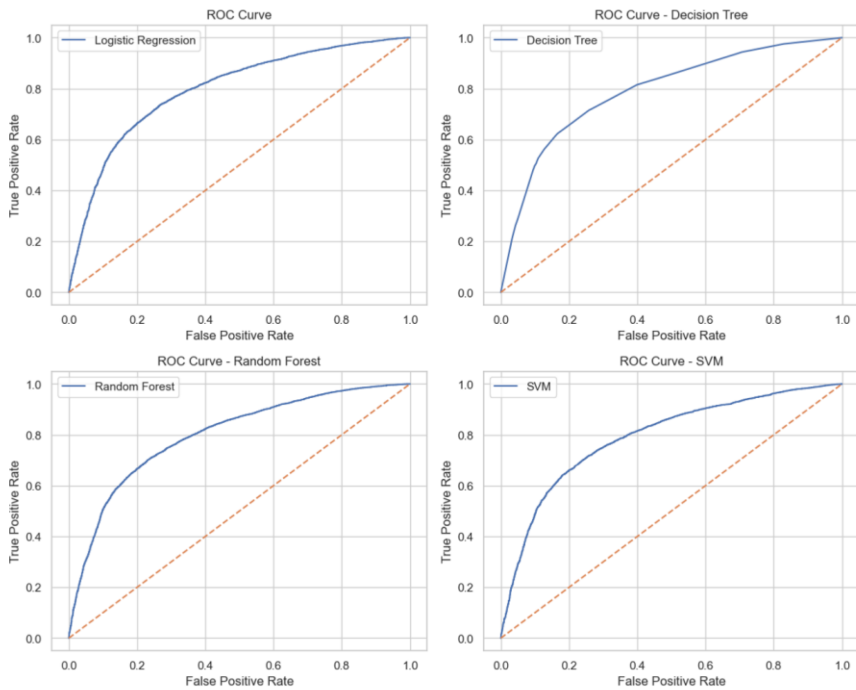

Figure 7. ROC curves for Logistic Regression, Decision Tree, Random Forest and Support Vector Machine.

The coefficient-based feature importance results showed that systolic blood pressure was the strongest predictor in the Logistic Regression model. Age, cholesterol, BMI and diastolic blood pressure also contributed strongly. Several interactions and quadratic terms were retained among the important predictors, including interactions involving blood pressure, age and cholesterol. This suggests that while the main predictor effects were strong, some combined and non-linear effects were also present in the dataset.

4.4 Decision Tree

The tuned Decision Tree model achieved an accuracy of 0.729, precision of 0.752, recall of 0.669, F1-score of 0.708 and AUC-ROC of 0.791 (Table 2). The confusion matrix showed 5,327 true negatives and 4,355 true positives, with 1,439 false positives and 2,152 false negatives (Fig. 6). Compared with Logistic Regression, the Decision Tree produced fewer false positives but more false negatives. This explains its relatively high precision but lower recall.

The ROC curve for the Decision Tree showed reasonable discriminatory performance, although it was slightly weaker than the Logistic Regression and Random Forest curves (Fig. 7). The curve remained clearly above the diagonal reference line, confirming that the model performed better than random classification.

The analyses showed that the Decision Tree relied heavily on systolic blood pressure, which accounted for approximately 77.9% of total feature importance. Age and cholesterol were the next most important predictors, while BMI, glucose and diastolic blood pressure contributed only small amounts. Gender and alcohol consumption did not contribute meaningfully to the final tree. The visualised tree also showed that systolic blood pressure and age appeared near the top of the decision structure, indicating that they were the main variables used to split patients into disease and non-disease groups.

4.5 Random Forest

The Random Forest model achieved the best overall predictive performance among the four models. It produced an accuracy of 0.734, precision of 0.757, recall of 0.673, F1-score of 0.713 and AUC-ROC of 0.799 (Table 2). The confusion matrix showed 5,359 true negatives and 4,382 true positives, along with 1,407 false positives and 2,125 false negatives (Fig. 6). This was the lowest number of false positives among all models, which contributed to the model's high precision.

The ROC curve for Random Forest showed the strongest overall discrimination, with the highest AUC-ROC value (Fig. 7). Although the difference between Random Forest and Logistic Regression was small, Random Forest performed slightly better in ranking patients by predicted cardiovascular disease risk.

Feature importance results confirmed that clinical variables dominated the prediction (Fig. 8). Systolic blood pressure (ap_high) was the most important predictor, accounting for approximately 47.7% of total importance. Diastolic blood pressure (ap_low) was the second most important predictor at 17.7%, followed by age at 13.7%, cholesterol at 8.8% and BMI at 6.4%. Together, these five predictors accounted for more than 94% of the model's total feature importance. In contrast, height, glucose, physical activity, gender, smoking and alcohol consumption made relatively small contributions.

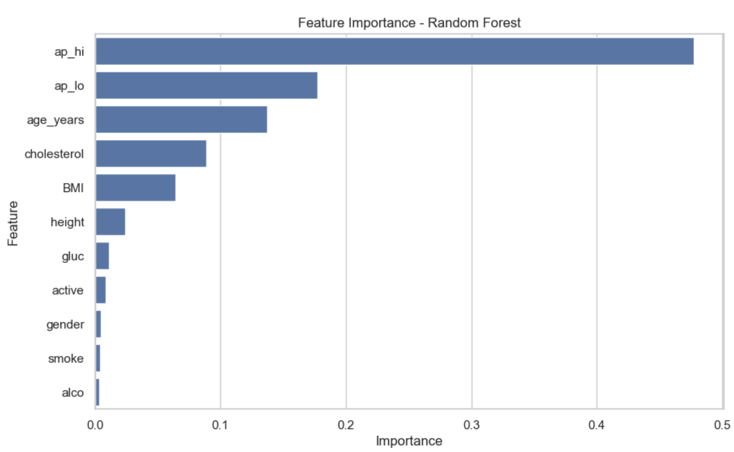

Figure 8. Feature importance plot for Random Forest. Note: Systolic Blood Pressure (ap_high) and Diastolic Blood Pressure (ap_low).

4.6 Support Vector Machine

The tuned SVM model achieved an accuracy of 0.731, precision of 0.756, recall of 0.664, F1-score of 0.707 and AUC-ROC of 0.791 (Table 2). The confusion matrix showed 5,374 true negatives and 4,319 true positives, with 1,392 false positives and 2,188 false negatives (Fig. 6). Similar to the Decision Tree, SVM achieved strong precision but lower recall, meaning that it was relatively conservative in predicting cardiovascular disease.

The ROC curve for SVM showed acceptable discriminatory performance and remained above the diagonal reference line across thresholds (Fig. 7). However, its AUC-ROC was slightly lower than both Random Forest and Logistic Regression. Although the RBF kernel allowed the SVM to model non-linear decision boundaries, this did not translate into a clear improvement in predictive performance.

4.7 Summary of Main Findings

Across all models, the strongest predictors of cardiovascular disease were systolic blood pressure, diastolic blood pressure, age, cholesterol and BMI. These variables were consistently identified as important in both coefficient-based and tree-based feature importance analyses. Lifestyle variables, including smoking, alcohol consumption and physical activity, showed comparatively weak predictive influence in this dataset.

Random Forest achieved the best overall performance, but its advantage over Logistic Regression was small. Logistic Regression remained highly competitive and provided the strongest recall and F1-score, while also offering greater interpretability. Decision Tree and SVM performed reasonably well but were slightly weaker, mainly because of lower recall. Overall, the results suggest that cardiovascular disease prediction in this dataset was driven primarily by a small group of clinical and demographic variables, particularly blood pressure, age, cholesterol and BMI.

5. Discussion

This study set out to answer five related research questions: (i) which variables contribute most strongly to cardiovascular disease prediction; (ii) whether the relationships between predictors and disease status are mainly linear or non-linear; (iii) how well different machine learning models predict cardiovascular disease; (iv) which model provides the best balance between predictive performance and interpretability; and (v) what the main findings imply for cardiovascular disease risk assessment. The results provide reasonably clear answers to each of these questions.

In relation to the first research question, the analysis showed that cardiovascular disease prediction in this dataset was driven mainly by a small group of clinical and demographic variables. Systolic blood pressure emerged as the most influential predictor across the models, followed by diastolic blood pressure, age, cholesterol, and BMI. This pattern was consistent in the Logistic Regression coefficients, the Decision Tree feature importance output, and the Random Forest feature importance plot. The dominance of blood pressure is clinically plausible because hypertension is a well-established risk factor for cardiovascular disease [1, 5]. Similarly, age, cholesterol, and BMI are widely recognised as important contributors to cardiovascular risk. Therefore, the modelling results support existing medical understanding while also confirming that these variables carried the strongest predictive signal in the selected dataset.

The second research question asked whether the relationships between predictors and cardiovascular disease were mainly linear or whether strong non-linear patterns and interactions were present. The results suggest that the dataset contained some non-linear and interaction effects, but these effects were not dominant (Fig. 8). This is supported by the close performance of Logistic Regression and Random Forest (Table 2). Logistic Regression, even though it is a simpler model, achieved performance almost equivalent to Random Forest after interaction terms, quadratic terms, and regularisation were included. If the dataset had contained strong and complex non-linear relationships, a larger performance gap in favour of Random Forest or SVM would have been expected. Instead, the small difference between models suggests that the main risk structure was largely captured by the major clinical predictors in an approximately linear or moderately non-linear way.

The third research question concerned the comparative predictive performance of the four machine learning models. Random Forest achieved the best overall performance, with the highest accuracy, precision, and AUC-ROC (Table 2). However, the difference between Random Forest and Logistic Regression was small. Logistic Regression achieved the strongest recall and F1-score, meaning that it performed slightly better in identifying actual disease cases. Decision Tree and SVM also performed reasonably well but were slightly weaker, particularly in terms of recall. This indicates that all four models were capable of learning meaningful patterns from the dataset, but Random Forest and Logistic Regression were the most useful overall.

The fourth research question focused on the balance between prediction and interpretability. Although Random Forest was the strongest model by a small margin, Logistic Regression offered a better balance between performance and transparency. This distinction is important in healthcare, where a model's usefulness depends not only on its accuracy but also on whether clinicians can understand and justify its predictions. Logistic Regression provides coefficients that can be interpreted in relation to individual risk factors, making it easier to explain why a patient may be classified as high risk. Random Forest, by contrast, can model more complex patterns but is less transparent because predictions are made through an ensemble of many decision trees. Given the small difference in performance, Logistic Regression may be preferable in situations where interpretability is a priority, while Random Forest may be more suitable where predictive accuracy is the main goal.

The fifth research question asked what the results imply for cardiovascular disease risk assessment. The findings suggest that routine clinical variables, especially blood pressure, age, cholesterol, and BMI, can provide useful information for identifying individuals at higher cardiovascular risk. The relatively weak contribution of smoking, alcohol consumption, and physical activity should be interpreted cautiously. These variables were recorded as simple binary indicators, which may not reflect the intensity, duration, or long-term history of exposure. For example, the dataset could not distinguish between occasional and heavy smoking, or between different levels of physical activity. Therefore, the low model importance of these variables does not mean that they are unimportant clinically; rather, it may reflect limited measurement detail in the dataset.

The model-specific results provide further insight into the strengths and weaknesses of each approach. The Decision Tree model was easy to interpret and produced clear decision rules, but it had lower recall and AUC-ROC than Logistic Regression and Random Forest (Fig. 7). This may reflect the tendency of single decision trees to rely heavily on a small number of dominant splits, particularly systolic blood pressure, and to miss more subtle patterns in the data. Random Forest improved on this by averaging predictions across many trees, reducing instability and producing the best overall performance. SVM also performed competitively, but its RBF kernel did not provide a clear advantage over simpler methods. This suggests that the additional complexity and lower interpretability of SVM were not justified for this dataset.

Overall, the results reinforce an important principle in applied healthcare machine learning: more complex models do not always produce substantially better outcomes. In this study, Random Forest was the best-performing model, but Logistic Regression was very close and easier to interpret. This finding supports the use of simpler, transparent models as strong baselines in clinical prediction tasks. It also shows that model selection should be based on both statistical performance and practical usefulness, particularly when the model is intended to support health-related decisions.

5.1 Limitations

Several limitations should be acknowledged. *First*, as the task was performed as a project assignment for a class, the level of diligence and comprehensive literature review expected in a full research study was outside the scope of the work due to time limitations. *Second*, the source of the data and its authenticity could not be independently verified, although the dataset is known as a recognised benchmark dataset available from Kaggle. *Third*, the dataset was cross-sectional, so the results identify associations and predictive patterns rather than causal relationships. *Fourth*, several clinically important variables, such as medication use, family history, ethnicity, diet, duration of smoking, and previous medical history, were not available. *Finally*, lifestyle variables were simplified into binary indicators, which may have underestimated their true contribution to cardiovascular disease risk.

5.2 Future Research Directions

Future work could improve the analysis by validating the models on external clinical datasets, incorporating more detailed medical and lifestyle variables, and applying additional modelling approaches such as Gradient Boosting, XGBoost, or calibrated ensemble models. Further research could also apply model explainability methods such as SHAP values to better understand how individual predictors influence patient-level predictions [7, 8]. These extensions would strengthen the clinical relevance of the analysis and provide a more robust basis for using machine learning in cardiovascular disease risk prediction.

6. Conclusion

This study applied machine learning techniques to predict cardiovascular disease using a benchmark dataset containing demographic, clinical, anthropometric and lifestyle-related variables. After data cleaning, feature engineering and preprocessing, four classification models were developed and compared: Logistic Regression, Decision Tree, Random Forest and Support Vector Machine. The analysis showed that all four models were able to predict cardiovascular disease with reasonable and broadly comparable performance.

Random Forest achieved the best overall result, with the highest accuracy, precision and AUC-ROC. However, the improvement over Logistic Regression was very small. This suggests that although Random Forest was slightly stronger as a predictive model, Logistic Regression remained highly competitive and offered the additional advantage of interpretability. In healthcare applications, where model transparency is important, this trade-off is particularly relevant.

The feature importance results showed that cardiovascular disease prediction was mainly driven by clinical and demographic variables. Systolic blood pressure was the strongest predictor, followed by diastolic blood pressure, age, cholesterol and BMI. These findings are consistent with established cardiovascular risk knowledge and indicate that routine health measurements can provide useful information for identifying individuals at higher risk. Lifestyle variables such as smoking, alcohol consumption and physical activity showed weaker predictive influence in this dataset, possibly because they were recorded only as binary indicators.

Overall, the study demonstrates that machine learning can support cardiovascular disease risk prediction, but model selection should consider both predictive accuracy and practical interpretability. More complex models may provide modest performance gains, but simpler models such as Logistic Regression may be more appropriate when clear explanation of risk factors is required. Future work should validate these findings using external clinical datasets and incorporate additional risk factors to improve predictive reliability and clinical relevance.

7. References

[1] H. E. Bays, P. R. Taub, E. Epstein, E. D. Michos, R. A. Ferraro, A. L. Bailey, H. M. Kelli, K. C. Ferdinand, M. R. Echols, and H. Weintraub, "Ten things to know about ten cardiovascular disease risk factors," *American journal of preventive cardiology,* vol. 5, pp. 100149, 2021.

[2] T. M. Powell-Wiley, P. Poirier, L. E. Burke, J.-P. Després, P. Gordon-Larsen, C. J. Lavie, S. A. Lear, C. E. Ndumele, I. J. Neeland, and P. Sanders, "Obesity and cardiovascular disease: a scientific statement from the American Heart Association," *Circulation,* vol. 143, no. 21, pp. e984-e1010, 2021.

[3] C. M. Bhatt, P. Patel, T. Ghetia, and P. L. Mazzeo, "Effective heart disease prediction using machine learning techniques," *Algorithms,* vol. 16, no. 2, pp. 88, 2023.

[4] T. Liu, A. Krentz, L. Lu, and V. Curcin, "Machine learning based prediction models for cardiovascular disease risk using electronic health records data: systematic review and meta-analysis," *European Heart Journal - Digital Health,* vol. 6, no. 1, pp. 7-22, 2025.

[5] G. I. Al Jowf, and M. Kolhar, "Key factors in predictive analysis of cardiovascular risks in public health," *Scientific Reports,* vol. 15, no. 1, pp. 23455, 2025.

[6] D. Han, "Cardiovascular disease predictive modeling with machine learning feature importance," *American Research Journal of Cardiovascular Diseases,* vol. 5, no. 1, pp. 1-6, 2023.

[7] S. M. Lundberg, and S.-I. Lee, "A unified approach to interpreting model predictions," *Advances in neural information processing systems,* vol. 30, 2017.

[8] A. V. Ponce-Bobadilla, V. Schmitt, C. S. Maier, S. Mensing, and S. Stodtmann, "Practical guide to SHAP analysis: Explaining supervised machine learning model predictions in drug development," *Clinical and translational science,* vol. 17, no. 11, pp. e70056, 2024.